



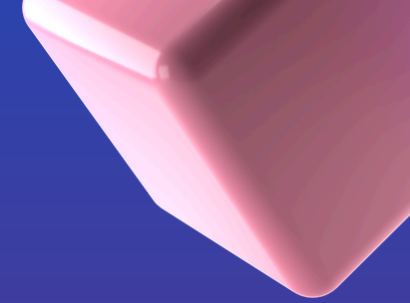
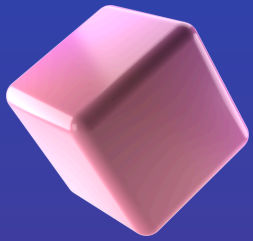
SECURITY & TECHNOLOGY

SECURING YOUR DIGITAL WORLD



THE DIGITAL WORLD TODAY

- Massive growth in cloud computing, mobile devices, and interconnected systems. Increased reliance on automation and digital infrastructure across industries



TYPES OF CYBER THREATS

Malware
(Viruses,
Worms,
Trojans)

Phishing

Ransomware

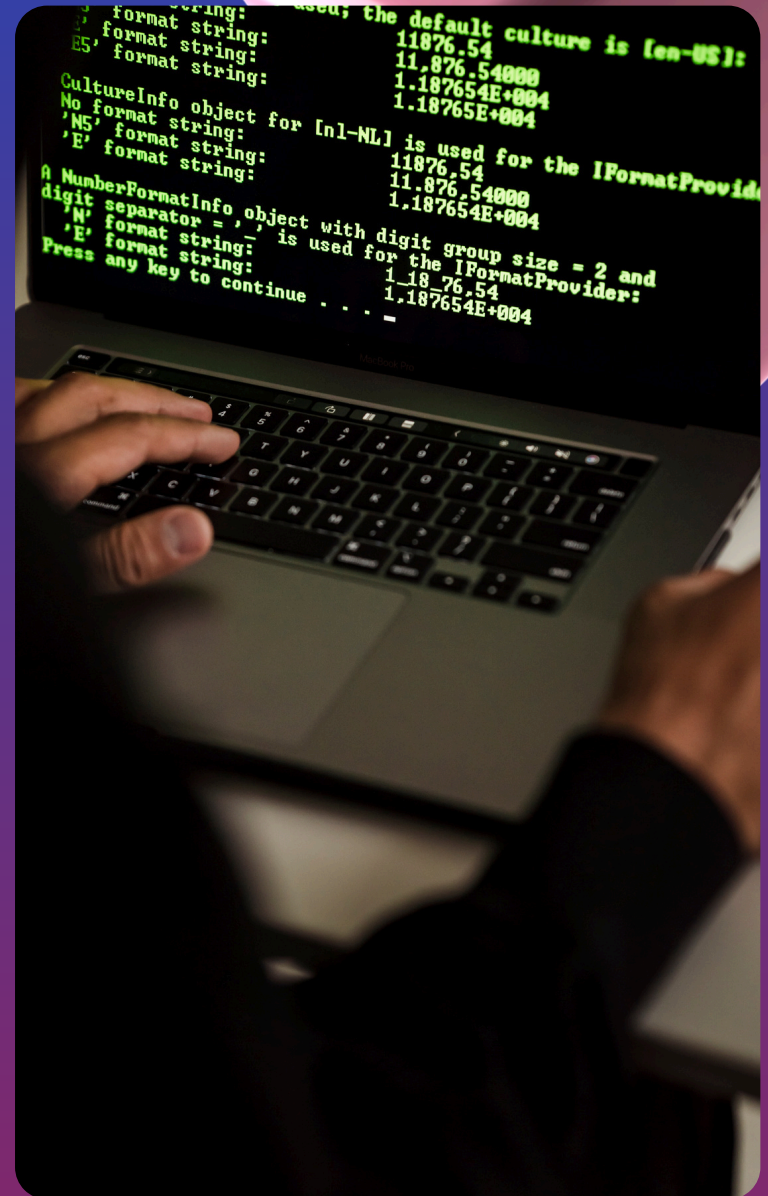
DDoS
(Distribut
ed Denial
of
Service)

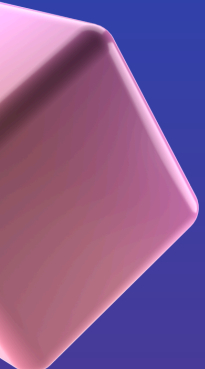
RANSOMWARE

Definition of Ransomware

malware that encrypts data and demands payment

- Attack vectors: phishing emails, compromised credentials, unpatched systems
- High-profile incidents affecting hospitals, schools, and government agencies
- How to Prevent: backups, segmentation, employee training, patching





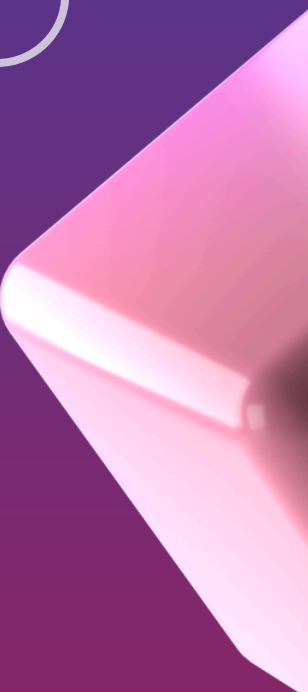
Shared responsibility
model: provider vs.
customer obligations

Common risks: misconfigured
storage buckets, weak access
controls, API
vulnerabilities

Importance of identity
and access management
(IAM)

Zero-trust
architecture as a
modern cloud security
standard

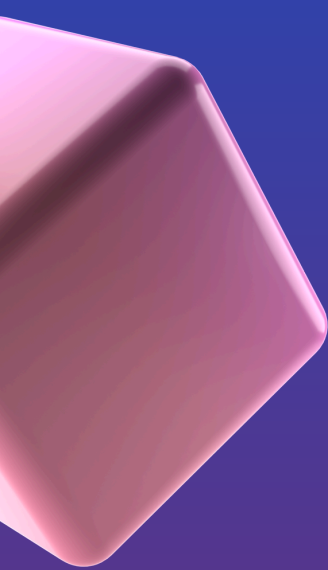
CLOUD SECURITY CHALLENGES



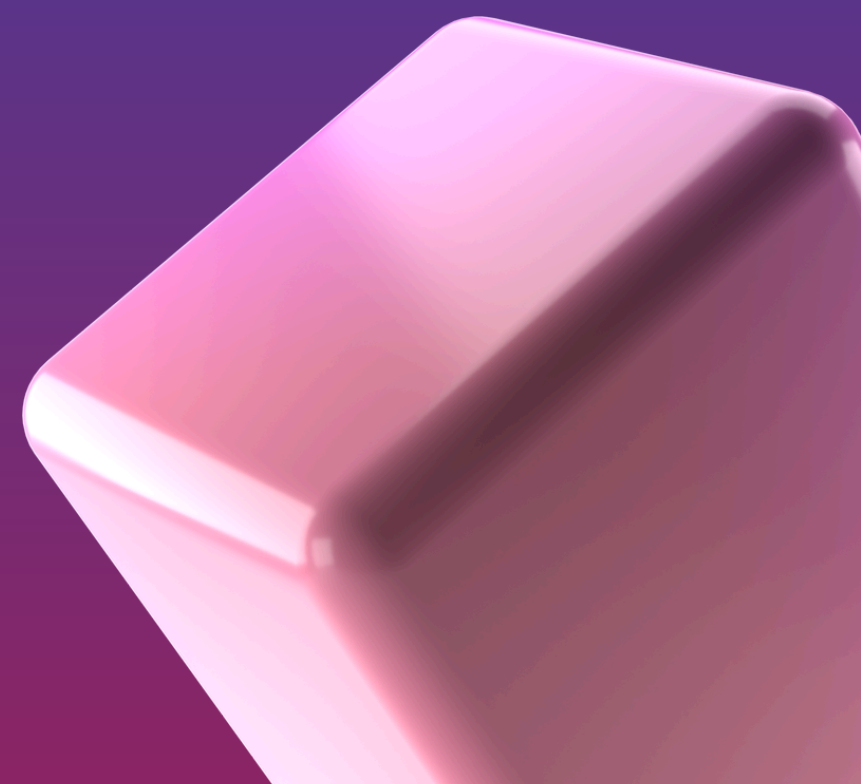
NETWORK SECURITY PRACTICES

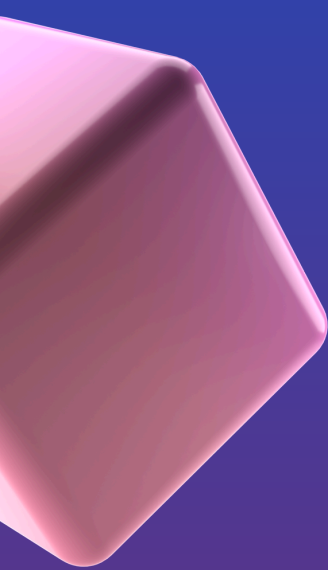
- Firewalls: filtering traffic and enforcing policies
- Intrusion Detection & Prevention Systems (IDS/IPS)
- Virtual Private Networks (VPNs) for secure remote access
- Network segmentation to limit lateral movement
- Importance of continuous monitoring and logging



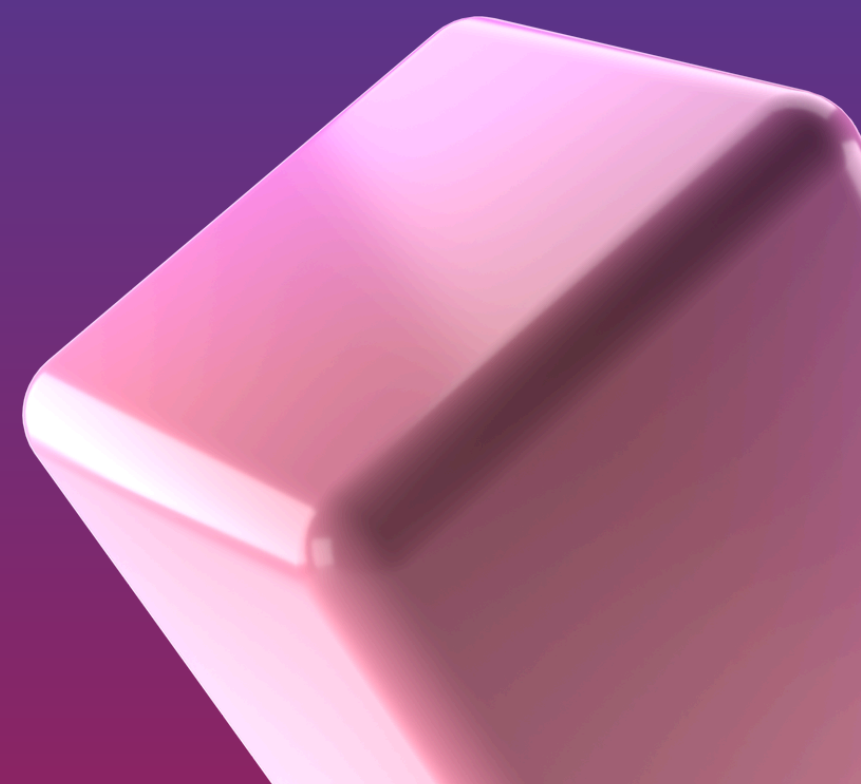


AI AND IT'S ROLE IN SECURITY

- AI for threat detection: anomaly detection, behavioral analytics
 - Machine learning models identifying patterns humans miss
 - Automated incident response and triage
 - Risks: adversarial attacks, model poisoning, deepfake-based social engineering
 - Ethical considerations and transparency in AI-driven security
- 



DATA PRIVACY & PROTECTION

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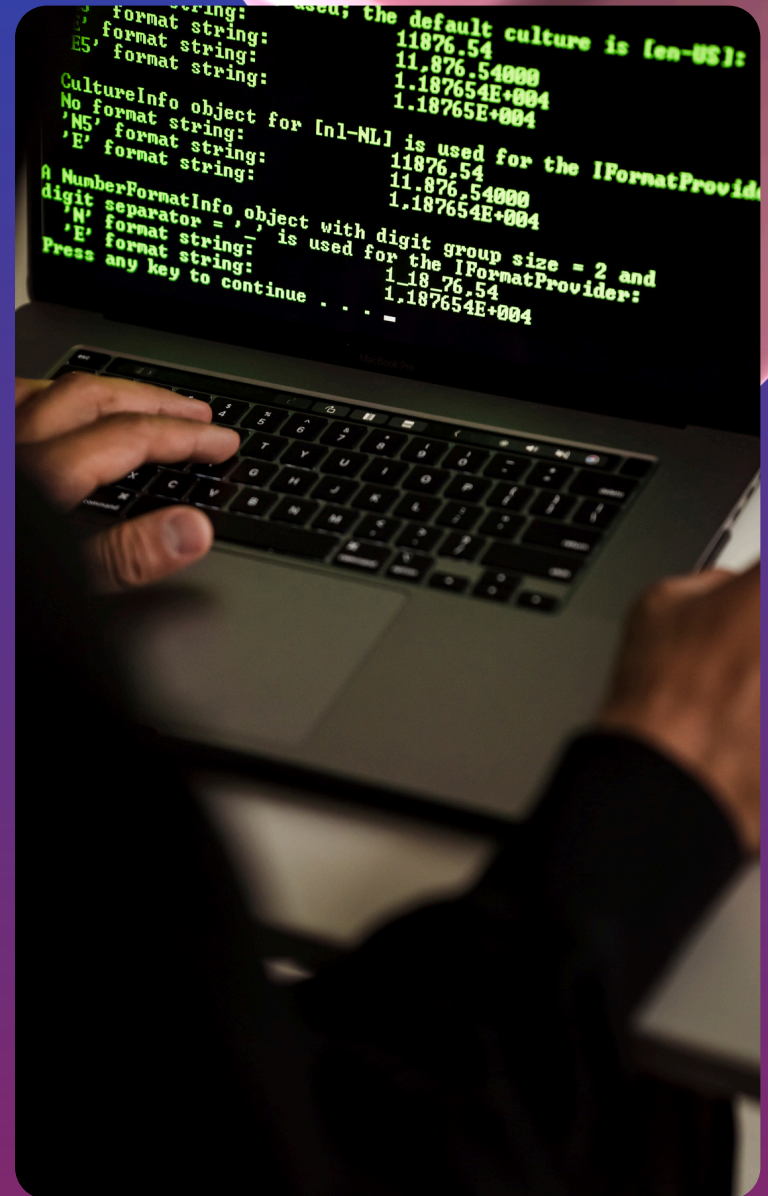


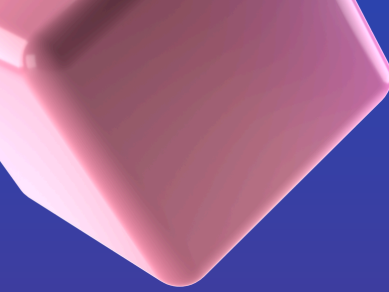
CONCLUSION

- Security must evolve alongside technology
- Organizations need layered defenses: technical, human, and procedural
- Proactive strategies outperform reactive responses
- The future of security depends on innovation, education, and resilience

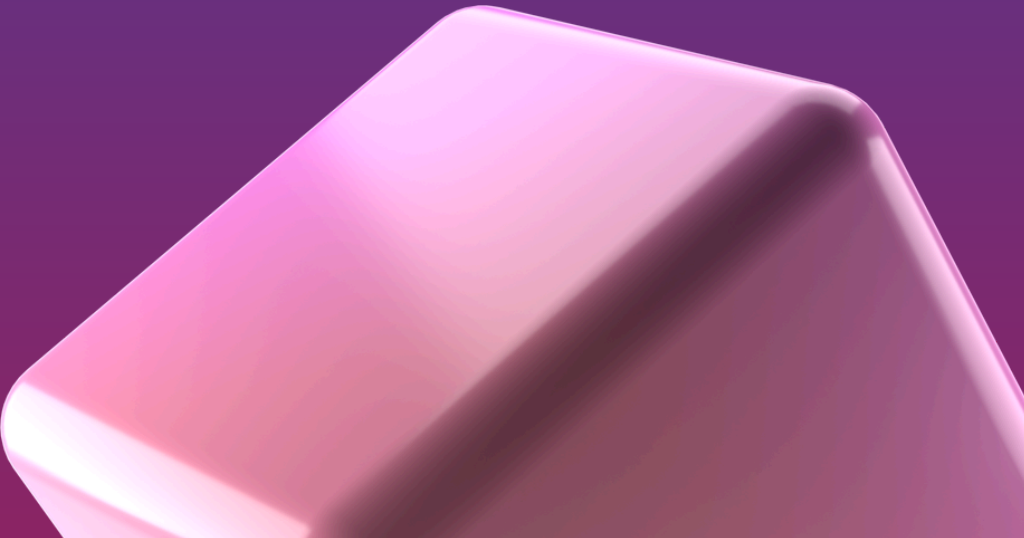
FUTURE OF SECURITY

- Quantum computing: potential to break current encryption
- Rise of bio metric authentication
- AI-powered cyberattacks becoming more sophisticated
- Autonomous systems and critical infrastructure vulnerabilities
- Increasing need for global cybersecurity collaboration





SOURCES

- <https://www.nist.gov/cyberframework>
 - <https://www.weforum.org/reports/global-cybersecurity-outlook-2024>.
 - <https://www.ibm.com/security/data-breach>
 - <https://www.ibm.com/think/topics/cloud-security>
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THANK YOU